

Report Using Wireshark

Task 5 – Capture and Analyze Network Traffic Using Wireshark

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Tool: Wireshark

Capture File: task5.pcapng

1. Objective

The objective of this task was to capture live network traffic using Wireshark, analyze the captured packets, identify different network protocols, and understand basic network communication.

2. Tools Used

- Wireshark

3. Procedure

Step 1: Opened Wireshark and selected the active network interface (eth0).

Step 2: Started packet capture.

Step 3: Generated traffic using:

- ping google.com
- Browsed <https://google.com>
- Browsed <https://github.com>

Step 4: Stopped the capture after approximately one minute.

Step 5: Applied Wireshark display filters:

- dns
- tcp
- icmp
- tls

Step 6: Exported the capture as task5.pcapng.

4. Protocols Identified

- DNS – Domain name resolution
- TCP – Reliable communication
- ICMP – Ping request/reply
- TLS – Encrypted HTTPS traffic
- ARP – IP to MAC address resolution

5. Findings

- Successfully captured live network traffic.
- Observed DNS queries generated during website access.
- Observed TCP connection establishment.
- Captured ICMP Echo Requests and Echo Replies.
- Identified encrypted TLS packets during HTTPS browsing.

6. Outcome

The task provided practical experience in packet capture, protocol filtering, and basic network traffic analysis using Wireshark.

7. Conclusion

Wireshark successfully captured and analyzed live network traffic. Multiple protocols were identified using protocol filters, demonstrating how packet analysis supports network troubleshooting and cybersecurity investigations.

References

- Wireshark Documentation
- Kali Linux Documentation
- TCP/IP Networking Fundamentals